# Manik Sharma

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## SUMMARY

Software engineer, Computer Science and Physics graduate (August 2026), who builds the systems that train and serve machine learning models, with PyTorch in hand. MIT Break Through Tech AI fellow and 1st of 240 teams in MIT's Equitable AI for Dermatology competition (76.3 weighted F1), with 200+ tracked experiments in W&B. Works in Python and PyTorch with a focus on making models reliable on real, messy data.

## EDUCATION

**University of Massachusetts Boston** Sept 2022 - Aug 2026

B.S. in Computer Science and Physics | GPA: 3.785

- Coursework: Machine Learning, Computer Graphics, Numerical Methods, Computational Physics, Compilers
- Awards: National Winner, Jo Anne J. Trow Scholarship (2025); Physics Departmental Award for Distinction (2026, sole undergraduate recipient in the major); Dean's List (7x); 2024 and 2025 Northrop Grumman Scholarship

## EXPERIENCE

**DOM Tech Academy** / *Software Engineer Co-op* Waltham, MA | Jun 2025 - Sep 2025

- Built and optimized features for an educational software platform (production code, deployments, debugging, API integrations); led instructional modules and mentored aspiring coders.

**Massachusetts Institute of Technology** / *Break Through Tech AI Fellow* Cambridge, MA | May 2024 - Apr 2025

- Led team to 1st place globally (240+ teams) in the Break Through Tech x Algorithmic Justice League Kaggle competition: CNN-to-ViT/Swin pipeline with phased training and weighted loss, 76.3 weighted F1 across 16 skin conditions and diverse skin tones.
- Executed 200+ hyperparameter sweeps via W&B Bayesian optimization and engineered targeted augmentation for underrepresented Fitzpatrick skin types to reduce model bias.

**Simulate Anything, Inc.** / *AI Studio Intern* Boston, MA | Aug 2024 - Dec 2024

- Developed a lightweight large action model for text-to-3D shape generation by composing geometric primitives via shape grammar, merging the ABC and DeepCAD datasets to redefine the grammar vocabulary.

## PROJECTS

**JobOps, AI-Native Application Automation** / *built end to end with AI coding agents* Jun 2026

- Built an automation system that scans 100+ job sources daily via ATS APIs, ranks postings with an explainable scoring model, and autofills forms in a browser for human review; every AI-generated component reviewed, tested, and corrected.

**Hear Me Out, AI Music Therapy Platform** / *2nd Place, HackHealth 2026 (ACM)* Mar 2026

- Built a five-stage Claude pipeline (narrative expansion, therapy-parameter mapping, lyrics with fidelity checking, arrangement, ElevenLabs song production) with 7-channel facial emotion tracking and crisis detection (988 Lifeline).

**Rizz The Grid, Energy Forecasting Platform** / *HackHarvard 2025* Oct 2025

- ML pipeline over 7 years of U.S. energy data (EIA API): ridge regression forecasting hourly CO2 intensity 14 days ahead, a PyTorch grid-stress optimizer, and anomaly detection on live streams; 50% higher forecasting accuracy than baselines.

## SKILLS

- **Machine learning:** PyTorch, TensorFlow, scikit-learn, pandas, W&B; computer vision, Transformers (ViT/Swin), NLP, time-series forecasting, anomaly detection, model evaluation, bias reduction
- **LLM and agents:** multi-stage Claude API pipelines, Gemini API, prompt engineering, LangChain, RAG, AI coding agents (Claude Code) with disciplined review of generated code
- **Software:** Python, JavaScript, TypeScript, C/C++, SQL, Node.js, FastAPI, Flask, MongoDB, SQLite, REST APIs, Git, UNIX/Linux